

Payroll Audit & Ghost Employee Detection using Excel XLOOKUP

Project Type: HR Payroll Reconciliation | Data Cleaning | Fraud Detection

Tools: Excel, XLOOKUP, COUNTIF, MAXIFS, SUMIFS, Power Query

Dataset: 500 Employees Master + 500 Haphazard Salary Records (Real-world messy data)

Excel Advanced Type Payroll Audit Status Completed

Business Problem

Company has two separate files:

1. **Master_Employee Sheet:** Clean list of 500 active employees (ID, Name, Designation)
2. **Salary_Haphazard Sheet:** 500 payroll entries - unsorted, with duplicates, extra IDs, and missing entries

Finance reported salary mismatch. Management suspected ghost employees.

My Task as Data Analyst: Reconcile both sheets, find who is not getting salary, and who is unknown entity drawing salary.

Dataset Overview

- **Master_Employee:** 500 rows - Employees_ID (EMP0001 to EMP0500), Employees_Name , Designation
- **Salary_Haphazard:** 500 rows - Employees_ID , Salary
 - Features of messy data:
 - Haphazard order (not same as Master)
 - 194 duplicate entries

- 30 rows with IDs not present in Master (EMP0501-EMP0530)
- 212 Master IDs missing in Salary sheet

 My Audit Findings

Metric	Count	Risk Level
Total Employees in Master	500	-
Total Salary Transactions	500 rows	-
Missing in Salary - Not Drawing Salary	212 employees (42.4%)	High - Compliance Risk
Fake / Extra IDs in Salary - Ghost Employees	18 Unique IDs / 30 Transactions	Critical - Fraud
Duplicate IDs Drawing Double Salary	130 IDs	High - Overpayment

Financial Impact (Estimated):

- Average Salary: ₹60,000
- Ghost Payroll: $30 * ₹60,000 = ₹18,00,000$ per month (~₹2.16 Cr / year) paid to unknown entity

 Methodology & Excel Formulas Used

Task 1: Basic Reconciliation with XLOOKUP

In Master sheet, find salary:

```
excel
```

```
1 =XLOOKUP(A2,Salary_Haphazard!$A$2:$A$501,Salary_Haphazard!$B$2:$B$501
```

Learning: Used \$ absolute reference + 4th argument for error handling instead of #N/A.

Task 2: Duplicate Detection

In Salary sheet:

excel

```
1 =IF(COUNTIF($A$2:A2,A2)>1,"Duplicate","Unique")
```

Found EMP0388 drawing ₹45,214 and ₹1,18,280 in same month.

Task 3: Ghost Employee Detection (Fake ID Check)

In Salary sheet:

excel

```
1 =XLOOKUP(A2,Master_Employee!$A$2:$A$501,Master_Employee!$A$2:$A$501,"
```

Flagged 18 IDs like EMP0515, EMP0401 which don't exist in Master.

Task 4: Find Missing Salary Drawers

In Master sheet:

excel

```
1 =MAXIFS(Salary_Haphazard!$B$2:$B$501, Salary_Haphazard!$A$2:$A$501, A2)
```

Used MAXIFS to handle duplicates and get latest salary.

Task 5: Financial Loss Calculation

excel

```
1 =SUMIF(D2:D501, "FAKE*", B2:B501)
```

Calculated total amount paid to ghost employees.



Final Deliverables Created

1. **00_Audit_Summary** - Executive summary for management
2. **01_Missing_IDs** - 212 IDs not drawing salary (to be verified for resignation / pending payment)
3. **02_Fake_or_Typo** - 18 ghost IDs (payment to be stopped immediately)
4. **03_Duplicate_Details** - 130 duplicate transactions (double payment check)
5. **Final_Report** - Clean master with Status column



Key Insights & Recommendations

1. **Immediate Action:** Stop payment for 18 fake IDs. Potential typo check: EMP0501 vs EMP0051.
2. **HR Master Cleanup:** 212 missing IDs - Verify if resigned. If active, release pending salary.
3. **Process Fix:** Implement XLOOKUP validation at data entry to prevent duplicate salary entry. Use Data Validation list for Employee ID.

4. **Future Audit:** Move this logic to Power BI with scheduled refresh to catch ghost employees monthly.

Why XLOOKUP, Not VLOOKUP?

- VLOOKUP cannot do reverse lookup (Name -> ID) - XLOOKUP can
- VLOOKUP breaks if column inserted - XLOOKUP with \$ doesn't
- VLOOKUP shows #N/A - XLOOKUP allows custom message "ID not in Salary Sheet"
- VLOOKUP can't handle duplicate handling with MAXIFS easily

Files in Repository

- `Employees_Task_2_Sheets.xlsx` - Raw messy data
- `Employee_Audit_Report.xlsx` - My final audit report with 5 sheets
- `README.md` - This file

About Me

Aspiring Data Analyst from Kolkata | Learning Power BI from Maven Analytics (Avi Singh) | Focus: Excel, DAX, Power Query

Connect: [\[LinkedIn\]](#) | [\[Portfolio\]](#)

This project demonstrates real-world payroll auditing skills beyond basic Excel formulas - handling duplicates, mismatches, and fraud detection using modern Excel.